DSO 429 · Digital Transformation of Business: AI and Smart Contracts

Homework 3

The Hourly Quiz

Assigned: Thursday, October 15, 2026
Due: Thursday, October 22, 2026, 2:00 PM
Submit via GitHub and Brightspace

For one week a quiz consisting of 25 multiple choice questions will go up every hour and you will have to submit your solutions to the questions on a public GitHub repo of your own. The first two days worth of quizzes are up on the questions repo and answers for these have been provided on the same repo to help you get the feeling of the type of solutions. From the third day, a new quiz is published every hour around the clock: 120 graded quizzes and 3,000 questions. You are free to come up with solutions to automate this.

The questions are deliberately easy, ordinary business arithmetic that a language model could help you answer correctly nearly every time. For example, here is question 1 of practice round 1:

> An investment of $31,500.00 grows at 12% compounded annually. What is its value after 7 years?
>
> A) $62,175.41 B) $57,960.00 C) $69,636.46 D) $26,460.00

The correct answer is C.

Schedule

	Dates	Rounds	What is published
Practice	October 15–17	1–48	All at once, with answer keys. Not graded.
Graded	October 17–22	49–168	One per hour. Questions only.

The 48 practice rounds and their answer keys are published before the assignment begins, so you can build and check your pipeline at your own pace. Use the practice days to break things; nothing before round 49 counts. Round 49 is released at 2:00 PM Pacific on Saturday, October 17, and a new round follows at the top of every hour until round 168 at 1:00 PM Pacific on Thursday, October 22. Leave your designed pipeline running through the practice days: the first new round it sees will be round 49. Timing is always measured from when a round was actually released, so a round published late never costs you time.

Part 1: Setup

S1 Repository

Create a public GitHub repository. A repository that has been deleted, renamed, transferred, or made private cannot be read, and every round in it is graded as missing.

S2 Registration

Submit: the repository's URL to the Brightspace assignment "Homework 3 GitHub Repo Link" by Friday, October 16, 11:59 PM, in the form:

```
https://github.com/<username>/<repository>
```

Part 2: Round Details

P1 Questions

Every round released so far is in one file, updated hourly once graded rounds begin:

```
https://github.com/FaisalXL/dso429-hourly-quiz/blob/main/questions.csv
```

GitHub can take up to five minutes to show a newly released round, so a round released at 2:00 PM may not appear in the file until 2:05 PM. The file has one line per question, with these columns:

Column	What it means
`round_id`	The round number.
`q_index`	The question's position in the round, 1–25.
`released_at`	When the round was released, in UTC.
`question_text`	The question.
`option_a`–`option_d`	The four choices. Exactly one is correct.

More than one round can be out at once, for example after a delay, so answer every round you have not answered yet, not just the newest. The answer keys for practice rounds 1–48 are at:

```
https://github.com/FaisalXL/dso429-hourly-quiz/blob/main/practice_keys.csv
```

P2 Commit the Answers

Commit one file per round to the `main` branch of your repository, with the round number padded to four digits:

```
answers/round-0001.json
```

The file holds all 25 answers. This is the correct file for practice round 1, so you can compare it with `practice_keys.csv`:

```
{
  "round_id": 1,
  "answers": {
    "q1":  "C", "q2":  "A", "q3":  "C", "q4":  "B", "q5":  "D",
    "q6":  "A", "q7":  "C", "q8":  "A", "q9":  "C", "q10": "A",
    "q11": "A", "q12": "D", "q13": "A", "q14": "B", "q15": "A",
    "q16": "D", "q17": "A", "q18": "B", "q19": "A", "q20": "C",
    "q21": "B", "q22": "D", "q23": "B", "q24": "B", "q25": "A"
  }
}
```

- `q1` through `q25` match `q_index`. Each value is `A`, `B`, `C`, or `D`; anything else is marked wrong.
- The file name decides the round, not the `round_id` inside it.
- A file that is missing, misnamed, or not valid JSON counts as not submitted.
- Several rounds may go in a single push.

After a few rounds, your repository looks like this:

```
<your repository>/
    answers/
        round-0001.json
        round-0002.json
        round-0003.json
        ...
```

P3 Push to GitHub

Make sure your answers are pushed to your public GitHub repository. That is what you are graded from.

Submit: one answer file per round, pushed to `main`.

Part 3: Usage Report

At the start of the assignment, update the usage tracking skill to its latest version. From your course root, the folder that contains all your assignment folders, run the installer once:

macOS

```
curl -O https://raw.githubusercontent.com/FaisalXL/claude-usage-tool/main/install.py
python3 install.py
```

Windows

```
curl.exe -O https://raw.githubusercontent.com/FaisalXL/claude-usage-tool/main/install.py
python install.py
```

After this it keeps itself up to date.

On the last day, Thursday, October 22, after 2:00 PM, generate the report in either of two ways:

- In Claude Code, run `/weekly-report` from anywhere inside your course folder.
- In a terminal, from your course root (on Windows, `python` instead of `python3`):

```
python3 .claude/weekly_report.py --days-back 7
```

Either way, it reports on the last seven days, which covers the whole assignment, and writes two files to `usage_reports/`: a computed summary, `<date>.json`, and the underlying session data, `<date>.transcript.jsonl`.

Submit: the `<date>.json` file only, to the Brightspace assignment "Homework 3". Keep the transcript on your machine.

Grading

Each graded round is scored as **fraction correct** × **timing credit**. Timing credit depends on how long after the round's release GitHub received your submission:

Hours after release	0–3	3–4	4–5	5–6	6–7	7–8	8–9	9–10	10–11	11–12	12+
Timing credit	100%	90%	80%	70%	60%	50%	40%	30%	20%	10%	0%

For example, 23 of 25 correct, received 5 hours 20 minutes after release, scores $0.92 \times 0.70 = 64.4\%$ for that round.

Which version counts. You may push a round's file as many times as you like.

- The last version received within three hours of release is graded.
- If nothing arrived within three hours, the first version received after that is graded, at its timing credit.

In practice: correct your answers freely for three hours. After that, a submitted answer is locked.

How time is measured. A submission's time is when its push reached your GitHub repository. Commit dates are not used, since they can be forged; we use the time GitHub records when each push comes in.

Final grade. The average of your 120 graded round scores, with your lowest six dropped. Rounds affected by a GitHub-wide outage confirmed on `githubstatus.com` are excused.

Submission Checklist

Item	**When**
Repository URL, Brightspace "Homework 3 GitHub Repo Link"	Friday, October 16, 11:59 PM
Practice rounds 1–48, in `answers/` on `main`	Any time, October 15–17
Graded rounds 49–168, in `answers/` on `main`	Hourly, October 17–22
Usage report `<date>.json`, Brightspace "Homework 3"	Thursday, October 22
Repository kept public, name and owner unchanged	Until grades are released

Questions

For any issues, reach out to `flnu@usc.edu`.